EXPERIMENT NO. 9
RESISTANCE OF GALVANOMETER BY KELVIN'S METHOD

Aim : To determine the resistance of a galvanometer by Kelvin's method using Wheatstone's meter bridge

Apparatus : Wheatstone's meter bridge, a rheostat, a cell, a galvanometer, a resistance box, a jockey, connecting wires, a cell or battery, a plug key etc.

Formula :

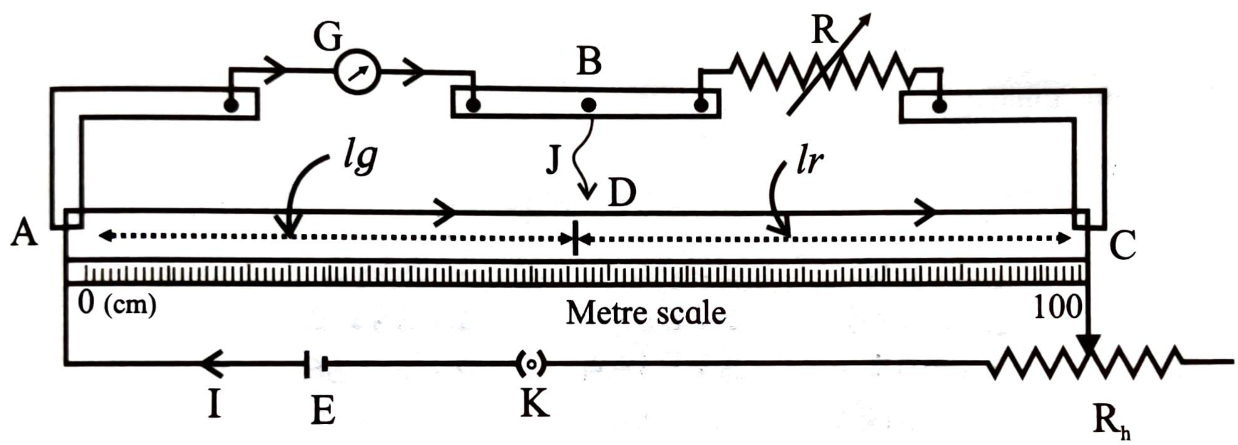
$$G = R\frac{\ell_G}{\ell_R}$$ where ℓ_G = length of the bridge wire corresponding to G

ℓ_R = length of the bridge wire corresponding to R

G = resistance of Galvanometer

R = resistance introduced in R.B

Circuit Diagram :



Procedure :

1. Connect the circuit as shown in figure
2. The galvanometer, whose resistance G is to be determined is connected in one arm and a resistance box is connected in the other arm of the Wheatstone's meterbridge. A jockey is directly connected from point B. A suitable resistance is introduced in the resistance box.
3. The circuit is closed and the galvanometer deflection is noted. The rheostat is adjusted so that the galvanometer shows nearly half the full scale deflection.
4. When the jockey is touched on the wire, the galvanometer deflection either increases or decreases. Move the jockey along the wire till the galvanometer deflection is restored to the original value. This is the null point or balance point. Thus in this position, the null point is constant when the jockey is touched or removed from the wire. Adjust R so that the balance point is between 30 cm and 70 cm, preferably one reading near or in the middle of the bridge wire. Measure ℓ_G and ℓ_R
5. Take three more readings by changing the values of R
6. Interchange the position of G and R and take four readings by adjusting R. calculate the value of G in each case. Find mean G.

Observation table :

1) Galvanometer in left gap

Obs. No.	R ohm	ℓ_G cm	ℓ_R cm	$G = R \frac{\ell_G}{\ell_R}$ ohm	Mean G ohm
1	50	44	56	33.28	61.23
2	70	46.7	53.3	61.33	
3	90	48	52	83.08	
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2) Galvanometer in right gap

Obs. No.	R ohm	ℓ_G cm	ℓ_R cm	$G = R \frac{\ell_G}{\ell_R}$ ohm	Mean G ohm
1	50	42.6	57.4	31.11	61.54.
2	70	47.4	52.6	63.06	
3	90	48.7	51.3	85.45	
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Calculations :

Log calculation table : $G = R \frac{\ell_G}{\ell_R}$

1) Galvanometer in left gap

Obs. No.	1	2	3	4
log R = a	1.6990	1.8451	1.9542	
log ℓ_G = b	1.6435	1.6693	1.6812	
a + b = c	3.3425	3.5144	3.6354	
log ℓ_R = d	1.7482	1.7267	1.7160	
c – d = e	1.5943	1.7877	1.9194	
antilog e = G	3.929×10^1	4.872×10^1	8.307×10^1	

2) Galvanometer in right gap

Obs. No.	1	2	3	4
log R = a	1.6990	1.8451	1.9542	
log ℓ_G = b	1.6294	1.6758	1.6875	
a + b = c	3.3284	3.5209	3.6417	
log ℓ_R = d	1.7589	1.7210	1.7101	
c – d = e	1.3695	1.7999	1.9316	
antilog e = G	3.711×10^1	6.305×10^1	8.542×10^1.	

∴ the resistance of the galvanometer

$$G = \frac{\text{mean G (of left gap)} + \text{mean G (of right gap)}}{2}$$

G = _______ Ω

Result :

The resistance of the galvanometer G = ______ Ω

Precautions :

1. Make all connections and the keys in the resistance box tight
2. Adjust the resistance R such that the point lies in the middle one third of the meterbridge wire or between 30 cm and 70 cm, preferably one reading near or on the mid of the bridge wire
3. Keep the circuit on only at the time of observation
4. Do not slide the jockey on the wire but tap it gently to get the exact null point

Questions

1. What is a balance point?

A balance point is the point on the potentiometer wire where the potential difference across the wire is equal to the emf of the cell and no current flows through the galvanometer. Hence, the galvanometer shows zero deflection.

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2. What will happen if the cell and galvanometer are interchanged?

If the cell and galvanometer are interchanged, the balance point remains unchanged, provided the galvanometer has sufficiently high resistance. The null point can still be obtained.

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3. Can you use a shunt for the galvanometer? Explain

Yes, a shunt can be used with a galvanometer. A shunt is a very low resistance connected in parallel with the galvanometer. It allows most of the current to pass through the shunt, protecting the galvanometer from large currents and converting it into an ammeter.

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Remark and sign of teacher :..

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